Web-Based Material Management Systems: A Literature Review and a Proposed Implementation

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***Abstract*—Material management systems help organizations track stock, issue materials and stay accountable for them. This paper reviews existing research on digital material tracking and inventory management and points out what is still missing. Studies on real-time tracking, ICT-based frameworks and RFID-assisted stock control are compared on their methods, advantages and limitations. The review shows that existing work concentrates mainly on construction logistics and on tracking hardware, while a lightweight web application that combines inventory records, material issue and return, role-based login, an administrative dashboard and reports in one place gets little attention. To address this gap, the paper presents the proposed capstone project: a web-based Material Management System built with HTML, CSS, JavaScript, Node.js, Express.js and SQLite. Functional testing on a local computer confirmed that login, material issue and material return worked. No performance or usability measurements have been collected yet, so no efficiency gains are claimed. The paper ends with directions for further evaluation.**

Keywords—material management, inventory management, web application, role-based access control, Node.js, literature review

I. INTRODUCTION

Organizations depend on equipment and materials to carry out their daily work. Accurate information about where materials are, who has taken them and when they were returned is needed for coordination, resource planning and accountability.

When material records are kept in registers, spreadsheets or separate applications, finding information and preparing reports becomes slow.

Managers may have to check several records to see the stock position, and store staff may have to match issue and return entries by hand.

Researchers have studied digital approaches to this problem, including real-time material tracking, ICT-based management frameworks and RFID-assisted stock control [1]–[3]. However, these studies differ from a small, general-purpose web application in their technology and in the environment where they were tested.

The aim of this paper is threefold: to summarize the existing work, to identify the research gap, and to show how our capstone project, a web-based Material Management System, addresses that gap. The rest of the paper is organized as follows. Section II reviews the literature, Section III compares the studies, Section IV states the research gap, Section V describes the proposed work, and Section VI concludes the paper.

II. LITERATURE REVIEW

This section summarizes each reviewed paper in terms of what the authors did, the technology they used and the limitation we found.

A. Real-Time Tracking for Construction Logistics

Zhao et al. [1] studied the use of real-time tracking of materials and labor to support kit-based logistics management in construction. Their framework connects material information with scheduling information, which shows the value of joining material flow data with planning data. The work is built around construction logistics, so it does not cover a general inventory system with issue and return records and user roles.

B. ICT-Enabled Materials Management

Kasim [2] discussed how information and communication technologies can improve material tracking and inventory management in complex projects, and proposed a framework for ICT-enabled materials management. The paper highlights the importance of structured digital records. Its limitation is that it is framework-oriented and was published in 2010, so it does not reflect current web technologies.

C. RFID-Aided Stock Management

Moon et al. [3] examined an RFID-aided tracking system for stock management in a construction supply chain, with a scaffold supplier as the case. RFID gives a strong form of material traceability. The limitation is that it needs tags and readers, which adds hardware cost and makes the approach unsuitable for small setups that only need software-based records.

[TO BE COMPLETED BY THE GROUP: Add the remaining papers here, one short paragraph each, so that at least 15–20 papers are covered (last 5–7 years preferred). For each paper write what the authors did, the technology used and the limitation, in your own words. Read every paper before you cite it.]

TABLE I

COMPARISON OF THE REVIEWED PAPERS

No.	Author, Year	Method / Technology	Advantage	Limitation
1	Zhao et al., 2021 [1]	Real-time tracking of materials and labor for kit-based logistics in construction	Links material flow with scheduling information	Built for construction logistics; no general issue/return or role-based access
2	Kasim, 2010 [2]	ICT-enabled framework for materials management in complex projects	Shows the value of structured digital records for tracking and inventory	Framework-level discussion; published over a decade ago
3	Moon et al., 2018 [3]	RFID-aided tracking system for stock management in a construction supply chain	Gives technology-assisted traceability of materials	Needs RFID tags and readers; focused on a scaffold supplier
4				
5				
6				
7				
8				

Rows 4 onwards (yellow) are to be filled with the remaining papers; add more rows as needed.

III. COMPARISON OF THE STUDIES

Table I (above) compares the reviewed papers by author, year, method or technology, advantage and limitation. The studies agree that structured digital records and tracking improve material handling. They differ in approach, from software frameworks [1], [2] to hardware-based identification [3].

IV. RESEARCH GAP

Among the papers reviewed so far, the following gaps were found.

1) Domain focus: The studies concentrate on construction logistics and supply chains. A general web application for organizational stock, issue and return is not covered.

2) Hardware dependence: Tracking with RFID [3] needs extra devices. A software-only solution that runs in a browser is less explored.

3) Lack of integration: The reviewed work does not combine inventory records, issue and return, role-based login, an administrative dashboard and reports in one lightweight application.

4) Limited evaluation: Practical systems are often described without clear, documented test outcomes for basic workflows.

V. PROPOSED WORK

A. Overview

To address the gap, our capstone project is a web-based Material Management System. It maintains material inventory records, records material issue and return, provides login with role-based access, offers an administrative dashboard, and generates reports and analytics from the stored records. It uses standard web technologies and a relational database. It does not include RFID, GPS tracking, biometric attendance or automated physical identification of materials.

B. Technologies and Architecture

HTML structures the pages, CSS handles the visual design and JavaScript supports client-side interaction. Node.js and Express.js process requests and application logic, and SQLite stores the data. The system follows a client-server architecture with three layers, as shown in Fig. 1: the user works in the browser, the Express.js backend processes the request, and the SQLite database stores or returns the data.

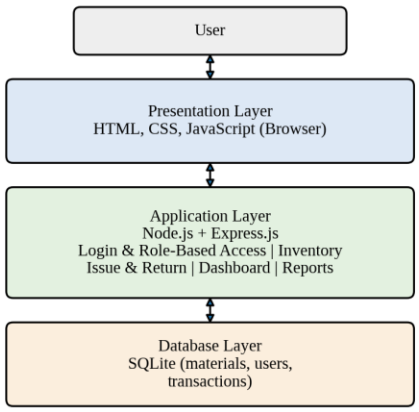


Fig. 1. Three-layer architecture of the Material Management System.

C. Modules

The material inventory module keeps material records. The issue and return module records materials taken from and given back to the store. The login module controls access, and role-based access limits functions according to user permissions. The dashboard gives administrators an overview of the available information, and the reports module presents information derived from stored records.

D. Development Method

The work followed six steps: requirement identification, frontend development, backend development, database integration, module integration and functional testing.

E. Functional Testing

The application was tested on a local computer. The confirmed outcomes are listed in Table II.

TABLE II

OBSERVED TEST OUTCOMES

Test case	Observed outcome	Status
Login	Login worked during testing	Passed
Material issue	Issue operation worked	Passed
Material return	Return operation worked	Passed

These results show that the tested workflows operated correctly in the local environment. Separate test results for the dashboard, reports and role-based access have not yet been recorded. Inventory accuracy, processing time, behavior with many users and user satisfaction have not been measured, so no improvement in productivity or efficiency is claimed.

F. Security Considerations

For a deployment-ready version, the system should use secure password hashing, server-side authorization checks, parameterized database queries, input validation, careful session handling and regular backups. These are recommendations and are not claimed as completed or security-tested features.

VI. CONCLUSION AND FUTURE SCOPE

Conclusion. This paper reviewed existing work on digital material tracking and inventory management and found that it concentrates on construction logistics and tracking hardware. A lightweight web application that integrates inventory, issue and return, role-based access, a dashboard and reports was identified as a gap. The proposed Material Management System addresses this gap, and functional testing confirmed that login, material issue and material return worked on a local computer.

Future Scope. Future work includes recording detailed test cases, testing the dashboard, reports and role permissions, measuring inventory accuracy against physical stock and the time taken for common transactions, testing with many users, and collecting user feedback. Barcode or QR-code scanning, notifications and additional analytics may also be considered. These are proposals, not features of the current version.

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[4]–[20] Add the remaining references here in IEEE style (only papers you have actually read).